

GEOMETRIC INTELLIGENCE ENGINE (GIE)

Milk Hill 2001 - Spiral-Arm Boundary / Five-Bit Phase Analysis

Working analytical record - 16 August 2026

1. Scope and controlling geometry

This record documents the current geometry-defined decoding work through the six-arm boundary test. It does not alter the accepted GIE reconstruction or claim an English-language message that has not yet been reproduced.

Controlling quantity	Value
Overall formation diameter	787 ft
Largest circle diameter	72 ft
Smallest circle diameter	40 in
Total circles	409
Structure	1 center + 6 arms x 68 circles
Arm separation	60 degrees

2. Binary extraction

Within each arm, the binary rule is: bit = 1 when the current circle diameter is greater than the previous circle diameter; otherwise bit = 0. With 68 circles on each arm, each arm contributes 67 diameter-to-diameter comparisons. Across six arms this yields $6 \times 67 = 402$ binary comparisons.

The 402-bit stream contains 80 complete five-bit groups (400 bits) plus 2 trailing bits.

3. Direct five-bit output

Direct output:

SFKBOJHWXFQIGLATVV<29>YCRYSLPAIL<27>JKIKNLEST<28><27>JFARWKP<27><28>QKJIDQOEADISIJ<28>ZVGNYPJ<27>IML<28> E

4. The arm-boundary phase result

The key structural result is arithmetic, not interpretive: 67 is not divisible by 5.

$$67 = 13 \times 5 + 2$$

Therefore, if five-bit grouping is kept continuous while traversal moves from one 68-circle arm to the next, the five-bit frame advances by 2 bits at every arm boundary. The starting phase evolves:

$$0 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 3 \rightarrow 0$$

5. Five-bit groups that actually cross arm boundaries

Boundary	Global bits	Five bits	Value	Interpretation
Arm 0 -> Arm 1	66-70	01100	12	L
Arm 1 -> Arm 2	131-135	00001	1	A
Arm 2 -> Arm 3	201-205	11011	27	CONTROL 27
Arm 3 -> Arm 4	266-270	01001	9	I
Arm 4 -> Arm 5	335/336	Boundary falls exactly between groups	-	No cross-boundary group

This shows that some five-bit symbols are literally composed from bits on two neighboring spiral arms. That makes the arm boundary a mathematically real part of the decoding problem.

6. Per-arm reset check

As a control test, each arm was also decoded independently by restarting five-bit grouping at the beginning of that arm. Each arm produces 13 full groups plus a 2-bit remainder:

Arm	13-group output	2-bit remainder
0	SFKBOJHWXFQIG	01
1	PFRZ<27>WDNKFMR SPACE	00
2	TVMUETUWFBYZN	01
3	VTLCENWAWYBVT	10
4	IDKYHJIBLZJRO	00
5	ZVGNYPJ<27>IML<28> SPACE E	10

7. Between-arm / radial-column test

To test the specific idea that information may be defined between neighboring spiral arms, the corresponding radial comparison at each rank was examined across all six arms. Direct six-bit columns, adjacent-arm XOR, and adjacent-arm equality/complement variants were tested. Under the existing alphabet/control mapping, these operations did not produce a reproducible English sentence.

That negative result is useful: it rules out several simple between-arm interpretations without changing the geometry or forcing language into the output.

8. Current result

- The 409-circle, six-arm geometry remains the controlling reconstruction.
- The binary extraction rule is reproducible and yields 402 comparisons.
- The direct five-bit stream is reproducible.
- The 2-bit phase advance at each 67-comparison arm boundary is an exact structural consequence.
- An English-language message has not yet been established by the tested boundary/radial-column rules.
- The next justified decoding work is to determine whether CONTROL values 27-31 have a geometry-defined role in handling arm-boundary phase/carry, rather than treating them as ordinary letters or arbitrarily rearranging output.

9. What Milk Hill shares with the world

What can be shared responsibly today is not a claimed sentence but a remarkable mathematical object: a large sixfold spiral that can be represented as a precise count, scale, radial ordering, binary comparison sequence, and phase structure. Its value is that it invites a reproducible question: can a visual geometric arrangement carry information under fixed rules?

The strongest public lesson is methodological. Geometry, measurement, binary encoding, and language must remain separate stages. A meaningful decode must survive a fixed rule without changing the rule after seeing the output. Whether or not the final stage becomes English, the formation already provides a concrete example of symmetry, sequence, information representation, and the difference between discovering a pattern and proving an interpretation.

10. Next test

Continue from the exact 2-bit arm-boundary phase shift. Test control values 27-31 specifically as possible boundary/carry instructions using only rules determined from the geometry and binary stream before reading resulting language. Preserve the existing direct stream and record every tested rule.

11. Control-code phase test

A constrained follow-up test treated five-bit values 27-31 as possible phase instructions because there are exactly five non-letter control values and five possible five-bit rotations. The natural mappings 27-31 -> phase positions 0-4 and 1-5 were tested as set, cumulative-add, and cumulative-subtract rules, with both left and right five-bit rotation. A parallel test treated the same values as small Caesar-style shifts on subsequent A-Z symbols.

None of these simple control-as-phase rules produced a stable English-language output. This means the control values cannot presently be assigned those meanings without adding an unsupported rule.

12. Direct two-arm carry symbols

A more geometry-specific boundary test was then performed. Each arm has 67 comparison bits = 13 complete five-bit groups + a two-bit remainder. For every neighboring arm pair, the two remainder bits from the first arm were joined to the first three comparison bits of the next arm. This creates exactly one five-bit symbol located at each arm-to-arm boundary. The sixth boundary closes Arm 5 back to Arm 0, respecting the sixfold cyclic arrangement.

Boundary	Carry 2	Next 3	Five bits	Value	Symbol
Arm 0 -> Arm 1	01	100	01100	12	L
Arm 1 -> Arm 2	00	101	00101	5	E
Arm 2 -> Arm 3	01	101	01101	13	M
Arm 3 -> Arm 4	10	010	10010	18	R
Arm 4 -> Arm 5	00	110	00110	6	F
Arm 5 -> Arm 0	10	100	10100	20	T

Boundary-symbol sequence: L E M R F T

This sequence is a real result of the specific two-arm carry construction. It is not yet an English word or sentence. No arbitrary rearrangement is accepted as a decode. The value of the result is that it produces one reproducible symbol for each of the six physical interfaces between neighboring arms.

13. Orientation controls

The same boundary construction was checked under the small set of geometry-defined orientation alternatives: clockwise versus counterclockwise arm order, outward versus inward traversal, and carry-bits-first versus next-arm-bits-first. The resulting six-symbol strings were:

- Clockwise / outward / carry first: LEMRFT
- Clockwise / outward / next first: QTUJXR
- Clockwise / inward / carry first: HLJIS<29>
- Clockwise / inward / next first: AQIENW
- Counterclockwise / outward / carry first: NREULD
- Counterclockwise / outward / next first: YJTVQP
- Counterclockwise / inward / carry first: KYRLHM
- Counterclockwise / inward / next first: MGJQAU

None is accepted as a linguistic decode. Several contain ordinary letter-like fragments by chance, but there is no geometry-defined reason to reorder their letters into a desired word.

14. Updated stopping point

The strongest new result is therefore not a sentence but a reproducible boundary alphabet candidate: six interfaces can each generate a five-bit symbol by carrying the two-bit remainder of one arm into the first three bits of its neighbor. The canonical clockwise/outward construction yields LEMRFT. Future work should test whether those six boundary symbols act as ordering keys, delimiters, or controls for the 13 complete five-bit groups inside each arm. Any such rule must be derived from the geometry before judging the resulting language.

15. Boundary-symbol selector test

The six boundary symbols L E M R F T have alphabet values 12, 5, 13, 18, 6, 20. Because each arm contains exactly 13 complete five-bit groups, the next constrained test used each boundary value as a selector into a 13-symbol arm block, reducing values greater than 13 modulo 13. Both possible boundary ownership conventions (current/outgoing arm versus next/incoming arm) and both radial directions (outward versus inward) were tested.

Boundary ownership	Radial direction	Six selected symbols
Current/outgoing arm	Outward	I <27> N E J <27>
Current/outgoing arm	Inward	F K T W B <27>
Next/incoming arm	Outward	R E T H J H
Next/incoming arm	Inward	F F V L I H

The most language-like result is RETHJH, which begins with RET, but the remaining symbols do not complete a stable English word and there is no geometry-defined reason to discard or rearrange them. It is therefore recorded as a negative/ambiguous result, not a decode.

16. Boundary-symbol ordering-key test

A second interpretation treated L E M R F T as an ordering key rather than as content. Sorting the boundary values numerically gives E, F, L, M, R, T (5, 6, 12, 13, 18, 20), corresponding to boundary order 1, 4, 0, 2, 3, 5. This ordering was used in two ways: concatenating whole 13-symbol arm blocks, and interleaving the six arms by radial group position. Both current-arm and next-arm ownership conventions were tested.

Current-arm block order: 1 - 4 - 0 - 2 - 3 - 5

Next-arm block order: 2 - 5 - 1 - 3 - 4 - 0

Neither concatenation nor radial interleaving produced a reproducible English sentence. The resulting streams retain the expected controls and letter-like fragments, but no complete message emerges under this ordering rule.

17. Boundary-key circular-start test

The boundary value was also used as a circular starting index within each 13-group arm block. This preserves every symbol and changes only the starting point. The four geometry-defined combinations tested were current-versus-next arm ownership and outward-versus-inward reading.

- Current / outward: IGSFKBOJHWFQ | <27>WDNKFMR PFRZ | NTVMUETUWFBYZ | ENWAWYBVTVTL C | JIBLZJROIDKYH | <27>IML<28> EZVGN YJ
- Next / outward: R PFRZ<27>WDNKF M | ETUWFBYZNTVMU | TVTLCENWAWYBV | HJIBLZJROIDKY | J<27>IML<28> EZVGN YJ | HWXFQIGSFKBOJ

No stable sentence appears. This test is therefore retained only as a documented elimination of a simple use of the six boundary letters as per-arm rotation keys.

18. Updated analytical conclusion

- The two-bit phase advance at each arm boundary remains the strongest exact decoding constraint.
- The six carry-derived boundary symbols LEMRFT are reproducible under the canonical clockwise/outward construction.
- Treating LEMRFT as direct text, selectors, arm-order keys, or circular-start keys has not produced a defensible English message.
- The tested simple meanings for control values 27-31 have also failed.
- No letters have been discarded, substituted, or manually rearranged to improve language.

The next justified test should work below the letter layer: use the five arm-boundary phase states themselves (0, 2, 4, 1, 3, 0) as a deterministic transposition or bit-alignment rule on the 13 complete groups in each arm. That test remains geometry-defined and does not require inventing a linguistic key.

19. Five-state phase alignment test

The exact arm-start five-bit phases are 0, 2, 4, 1, 3, 0 for Arms 0-5. These values were next applied below the letter layer, as required by the prior stopping point. Three deterministic uses were tested: (1) circularly shifting each arm's raw bit sequence by its phase before grouping, (2) rotating the bits inside each five-bit group by the arm phase, and (3) circularly shifting the 13 complete groups of each arm by the arm phase. Left/right variants were retained where the geometry permits both orientations.

None of these phase-state operations produced a stable English sentence. One output contains the fragment NCOREMI, which also appears in earlier permutation records; because it is not an independent complete linguistic result, it is recorded as a fragment only and is not promoted to a decode.

Representative raw-bit phase-left result: SFKBOJHWFQIG | ZKKN<28>QYLYVH | KFZRZJ<27>SAL<29>GF | MHXFJ<29>NCOREMI | IB<30>JBRHSFRTSX | ZVGN YJ<27>IML<28> E

20. Closed 402-bit ring test

The six-arm formation is cyclic, while the direct binary record is normally read as a linear 402-bit stream. A closed-ring test therefore joined the end of Arm 5 back to the beginning of Arm 0 and continued taking five-bit groups with a fixed stride of five bits, allowing groups to wrap across the end/start boundary.

This immediately reproduces the Arm 5 -> Arm 0 carry symbol T: the final two bits of Arm 5 are 10 and the first three bits of Arm 0 are 100, giving $10100 = 20 = T$. Thus the T in LEMRFT is also the first wrapped five-bit symbol of the closed 402-bit stream.

Number-theoretic property: $\gcd(402, 5) = 1$.

Because 402 and 5 are coprime, repeatedly advancing five bits around the closed 402-bit ring visits every possible bit-start position exactly once before returning to the original start. The resulting traversal contains 402 five-bit symbols and is therefore a complete cyclic scan of the binary ring rather than a hand-selected set of offsets.

The beginning of that cyclic scan reproduces the known direct stream, then wraps through T and continues into the next phase. Examination of the full 402-symbol cycle did not reveal a stable unambiguous English sentence under the existing 0=SPACE, 1-26=A-Z, 27-31=CONTROL mapping. The closed-ring construction is nevertheless retained because it is fully geometry-compatible and eliminates the arbitrary decision to stop decoding at the end of Arm 5.

21. New stopping point

- The six arm-boundary carry symbols LEMRFT remain reproducible.
- The final symbol T is independently reproduced by closing the 402-bit stream from Arm 5 back to Arm 0.
- The five-state phase sequence has been tested as raw-bit alignment, within-group rotation, and group-order rotation without a stable English result.
- The complete closed-ring five-bit traversal is mathematically unique once direction and starting point are fixed because $\gcd(402, 5) = 1$.
- No tested operation justifies deleting, substituting, or manually rearranging symbols.

The next defensible question is whether the geometry itself fixes the starting point and direction on the closed ring. If it does, the cyclic 402-symbol traversal becomes a fully specified decoder candidate; if it does not, language claims remain underdetermined.